Def. Let V be a vector space over a field F.
A function $f: V \times V \to F$ is called a bilinear form if: for all $\alpha_1, \alpha_2, \beta_1, \beta_2 \in V$, $c, d \in F$,

$$f(c\cdot\alpha_1 + \alpha_2, d\cdot\beta_1 + \beta_2) = cd\cdot f(\alpha_1, \beta_1) + c\cdot f(\alpha_1, \beta_2) + d\cdot f(\alpha_2, \beta_1) + f(\alpha_2, \beta_2).$$

Def. Let V be a fin. dim vector space over F, and let $\beta = \{\alpha_1, \cdots, \alpha_n\}$ be a basis for V.
Given a bilinear form f, define the matrix of f in β as;

$$[f]_\beta \overset{\text{def}}{=} (f(\alpha_i, \alpha_j))_{i,j}.$$

Thm. Let V be a finite-dimensional vector space over F, and β be a basis.
Then, the space $L(V, V, F) = \{f: V\times V \to F \mid f \text{ is bilinear form}\}$ is isomorphic to $M_{n\times n}(F)$.
Proof. Define $\varphi: L(V,V,F) \to M_{n\times n}(F);\ f \mapsto [f]_\beta$. Denote $\beta = \{\alpha_1, \cdots, \alpha_n\}$.
Given $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ and $\beta = y_1\alpha_1 + \cdots + y_n\alpha_n$,

$$f(\alpha, \beta) = \sum_{i=1}^{n} x_i \cdot f(\alpha_i, \beta) = \sum_{i=1}^{n} x_i \cdot \sum_{j=1}^{n} y_j \cdot f(\alpha_i, \alpha_j) = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j \cdot f(\alpha_i, \alpha_j).$$

Thus, for $X = [\alpha]_\beta$ and $Y = [\beta]_\beta$, and $A = [f]_\beta$,

$$f(\alpha, \beta) = X^t A Y = [\alpha]_\beta^t [f]_\beta [\beta]_\beta.$$

Therefore, the bilinear form f is uniquely determined by the representation $[f]_\beta$.
This proves the φ is bijection. And, the φ is linear, because

$$(c\cdot f + g)(\alpha_i, \alpha_j) = c\cdot f(\alpha_i, \alpha_j) + g(\alpha_i, \alpha_j), \text{ for any } c \in F \text{ and } f, g \in L(V, V, F).$$

Cor. If $\beta = \{\alpha_1, \cdots, \alpha_n\}$ is a basis for V, and $\beta^* = \{f_1, \cdots, f_n\}$ is the dual basis, then $f_{ij}(\alpha, \beta) \overset{\text{def}}{=} f_i(\alpha)\cdot f_j(\beta)$ $(1 \le i, j \le n)$ form a basis for V^{**}.